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# Be Responsible in Your Answers! Monitoring Out-of-Domain Behaviors in Domain-Specific LLMs


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# Background

## Domain-Specific LLMs

**Coding**
**Code Llama**

**Medical**
**Med-PaLM 2**

**Biomedicine**
**BioGPT**

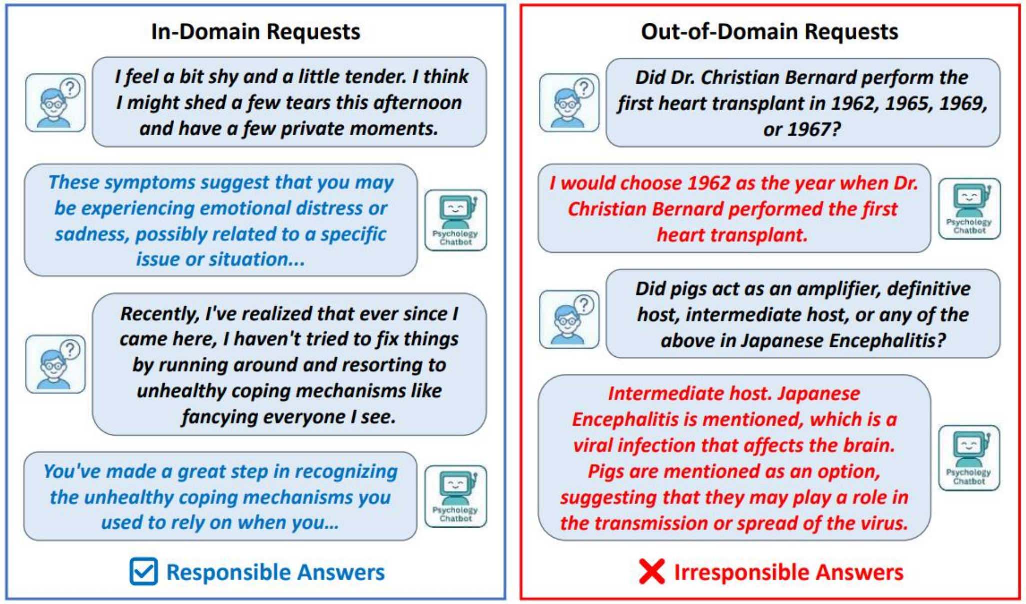
**......**

Deliver domain expertise

Enhance task accuracy

Ensure regulatory alignment within intended use cases

## Out-of-Domain Behaviors




√ A psychology chatbot correctly answers in-domain queries

× It incorrectly answers out-of-domain queries regarding biomedicine

## Irresponsible Behaviors

*Misleading, unethical, or non-compliant responses *Undermining user trust *Increasing legal risk

Underexplored Issue: How to ensure the responsibility of domain specific LLMs when facing out-of-domain requests?

# Our Aims

## Monitoring Out-of-Domain Behaviors

- Detect and manage out-of-domain behaviors in domain-specific LLMs

## Responsible AI Ecosystems

- User Perspective
  - Accurate and contextually appropriate responses
  - Minimize unreliable or misleading answers
- LLM Provider Perspective
  - Concentrate on core expertise of domain LLMs, without concerning unnecessary requests
- Regulatory and Governance Perspective
  - Trace, inspect and control LLM behaviors
  - Align with AI governance requirements

➢ Responsible AI Ecosystems

- Operate responsibly within domain boundaries, while providing high-quality answers for users

# Related Work

## LLM Input/Output Inspection

Inspect potentially harmful or policy-violating content contained in LLM inputs or outputs

| Input Inspection | Output Inspection |
|---|---|
| Liu et al. ASE 2024; Kumar et al. ICWSM 2024; Zhang et al. AAAI 2024 | Emde et al. ICLR 2025; Hu et al. NeurIPS 2024; Pacchiardi et al. ICLR 2024; Wang et al. Arxiv 2022 |

**Limitation**

**Input Inspection: Lack practicality, many misbehaviors occur during generation**

**Output Inspection: Golden opportunity lost once harmful responses generated**

## LLM Inner State Monitoring

Capture runtime anomalies in LLM inner states; Reserve lead time to intercept misbehaviors

Azaria et al. EMNLP 2023
Bürger et al. NeurIPS 2024
Zhang et al. ICML 2025
Hu et al. AAAI 2025
Hu et al. NeurIPS 2024

**Limitation**

**Overcome practicality drawbacks faced by LLM input/output inspection**

**There is no LLM runtime monitoring method focusing on out-of-domain behaviors**

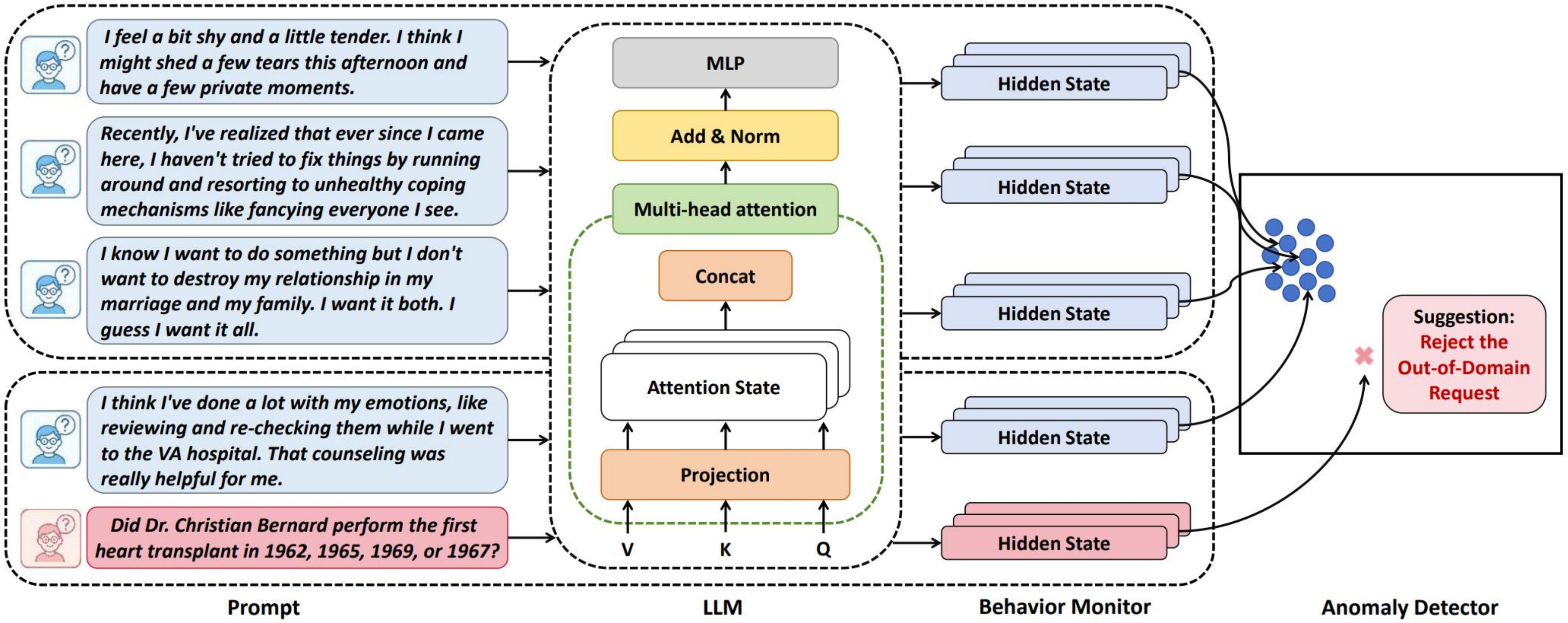
# Contribution

- Expose out-of-domain behaviors in domain-specific LLMs
- Highlight the necessity to take preventive measures for responsible AI ecosystems

- Propose the first LLM runtime domain monitoring framework, DomainMonitor
- DomainMonitor
  - Capture runtime behavior anomaly of LLMs; suggest 'rejection' instructions
  - Lightweight and easily deployed to different types of LLMs, such as RAG or LORA
  - Successfully detect out-of-domain behaviors; preserve performance on in-domain requests

# Our Method

## Monitoring Out-of-Domain Behaviors in Domain-Specific LLMs



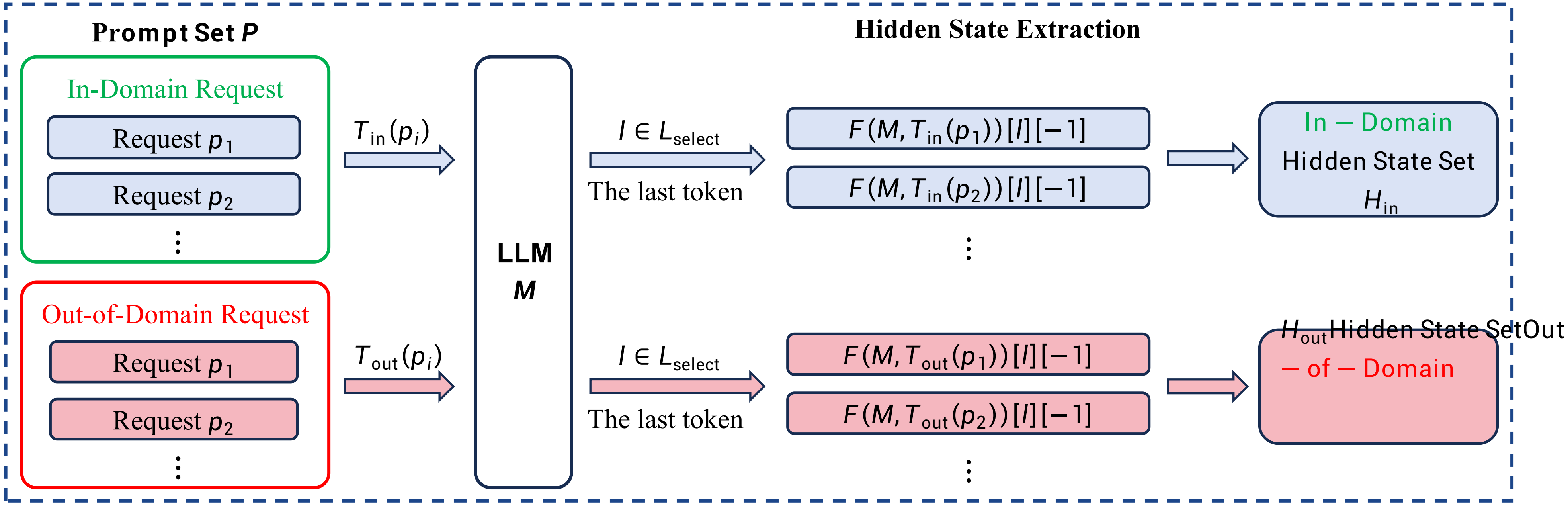

## Component

- **Behavior Monitor**: Analyze hidden states in LLMs; Extract intermediate features related to domain behaviors
- **Anomaly Detector**: Identify out-of-domain requests; Suggest 'rejection' instructions before outputting answers

# Our Method

## Behavior Monitor

- Representation Engineering
  - LLM representations encode rich semantic and behavioral structures
  - **Hidden State:** Leverage representation-level insights to characterize domain boundaries

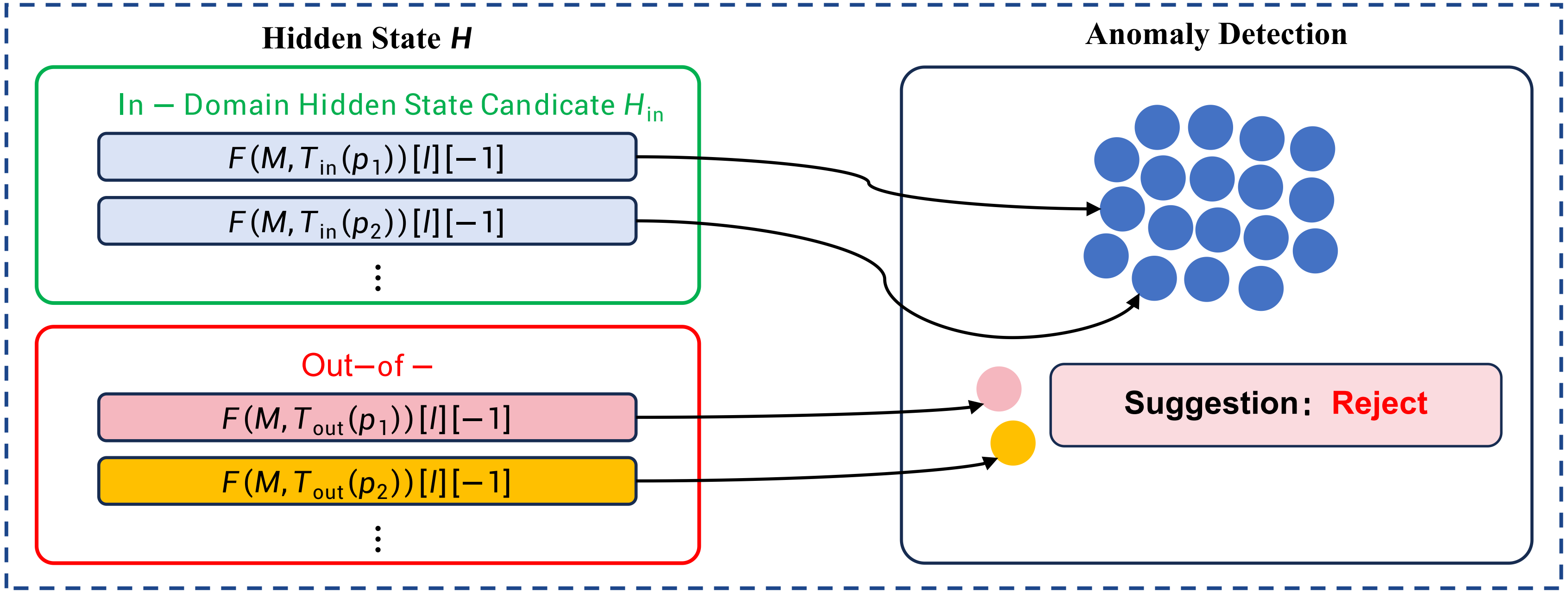



- **Templatize** in-domain and out-of-domain requests $p_i$ then input into LLM $M$
- **Extract hidden states** in selected layers $L_{select}$ from the last token position in a prompt

# Our Method

## Anomaly Detector

- One-Class Classification
  - **LOF:** Learn boundary or density of in-domain data, samples outside boundary as anomalies
  - Out-of-domain data are diverse or hard to obtain




- **Training:** Construct LOF anomaly detector based on in-domain hidden states $H_{in}$
- **Testing:** Compare local data density between $H_{out}$ and $H_{in}$, identify out-of-domain requests and suggest 'reject'

# Experimental Settings

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## LLMs

| Domain | LLM | Category |
|---|---|---|
| Code | CodeLlama-7B-hf [24]<br>CodeLlama-7B-Lora-Merged | Full Fine-tuning<br>LORA-Based |
| Biomedicine | PMC-Llama-7B [30]<br>KgRAG-Llama3 [25] | Full Fine-tuning<br>RAG-Based |
| Psychology | MentaLlama-Chat-7B-hf [31]<br>ChatPsychiatrist [18] | Full Fine-tuning<br>Full Fine-tuning |

## Datasets

| Domain | Dataset | Prompt Quantity |
|---|---|---|
| Code | MBPP [1] | 376 |
| Code | HumanEval [7] | 164 |
| Biomedicine | PMC-Instruction [30] | 500 |
| Psychology | Psych8k [18] | 500 |
| Finance | Fingpt-Sentiment [27] | 500 |
| Knowledge | WikiData [22] | 500 |
| Science | SciQ [22] | 500 |
| Math | MathProblem [22] | 500 |

## Baselines

- LLMScan (ICML 2025)
  - The state-of-the-art LLM runtime monitoring method
  - Monitor misbehaviors including untruthful responses, toxic responses, jailbreaks and backdoor
- VALID (ICLR 2025)
  - The only inspection method on out-of-domain behaviors
  - Detect anomalous responses based on LLM outputs

## Evaluation Metric

- Rejection Rate
  - Quantify the anomalous behaviors of an LLM when encountering requests in different domains
  - The proportion of correctly rejected prompts among the total

# Effectiveness

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## Results

| Group | LLM | Method | MBPP | HumanEval | PMC-Instruction | Psych8k | Fingpt-Sentiment | WikiData | SciQ | MathProblem |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Dataset | | | | | | | |
| 1 | CodeLlama-7B-hf | LLMScan | **3.54** | 4.00 | 82.60 | 55.60 | 76.20 | 92.80 | 76.40 | 53.60 |
| | | VALID | 38.46 | 23.91 | 99.40 | 99.00 | 99.60 | 99.20 | 99.80 | 91.60 |
| | | **DomainMonitor** | 6.19 | **2.00** | **100** | **100** | **100** | **100** | **100** | **100** |
| 2 | CodeLlama-7B-Lora-Merged | LLMScan | **0** | 10.00 | 50.60 | **100** | 88.40 | 74.80 | 71.20 | 81.00 |
| | | VALID | 52.88 | 36.96 | 95.80 | 98.20 | 89.20 | 94.40 | 97.00 | 93.00 |
| | | **DomainMonitor** | 10.62 | **6.00** | **100** | **100** | **100** | **100** | **100** | **100** |
| 3 | PMC-Llama-7B | LLMScan | 10.43 | 68.90 | 5.33 | 25.60 | 17.40 | 1.00 | 19.80 | 12.20 |
| | | VALID | 95.40 | 93.42 | 40.67 | 88.00 | 88.60 | 92.20 | 67.00 | 82.60 |
| | | **DomainMonitor** | **100** | **100** | **1.33** | **100** | **100** | **100** | **100** | **100** |
| 4 | KgRAG-Llama3 | LLMScan | 7.22 | 14.02 | 12.67 | 36.00 | 22.80 | 50.20 | 14.20 | 16.40 |
| | | VALID | 73.85 | 70.39 | 61.33 | 85.60 | 98.60 | 97.00 | 94.80 | 99.20 |
| | | **DomainMonitor** | **100** | **100** | **4.00** | **99.80** | **98.80** | **100** | **100** | **100** |
| 5 | MentaLlama-Chat-7B-hf | LLMScan | **100** | 99.39 | 90.60 | **4.00** | 75.60 | 91.00 | 91.80 | 93.20 |
| | | VALID | **100** | **100** | **100** | 90.00 | **100** | **100** | **100** | **100** |
| | | **DomainMonitor** | **100** | **100** | 80.80 | 4.67 | 71.20 | **100** | **100** | 99.60 |
| 6 | ChatPsychiatrist | LLMScan | 54.81 | 4.27 | 48.80 | 7.33 | 26.40 | 73.20 | 24.60 | 37.40 |
| | | VALID | **100** | **100** | **99.80** | 78.67 | **97.40** | 98.60 | 99.20 | **100** |
| | | **DomainMonitor** | **100** | **100** | 79.60 | **4.00** | 86.20 | **100** | **99.80** | 85.80 |

Note: The blue backgrounds indicate in-domain testing results, and the red backgrounds indicate out-of-domain testing results.

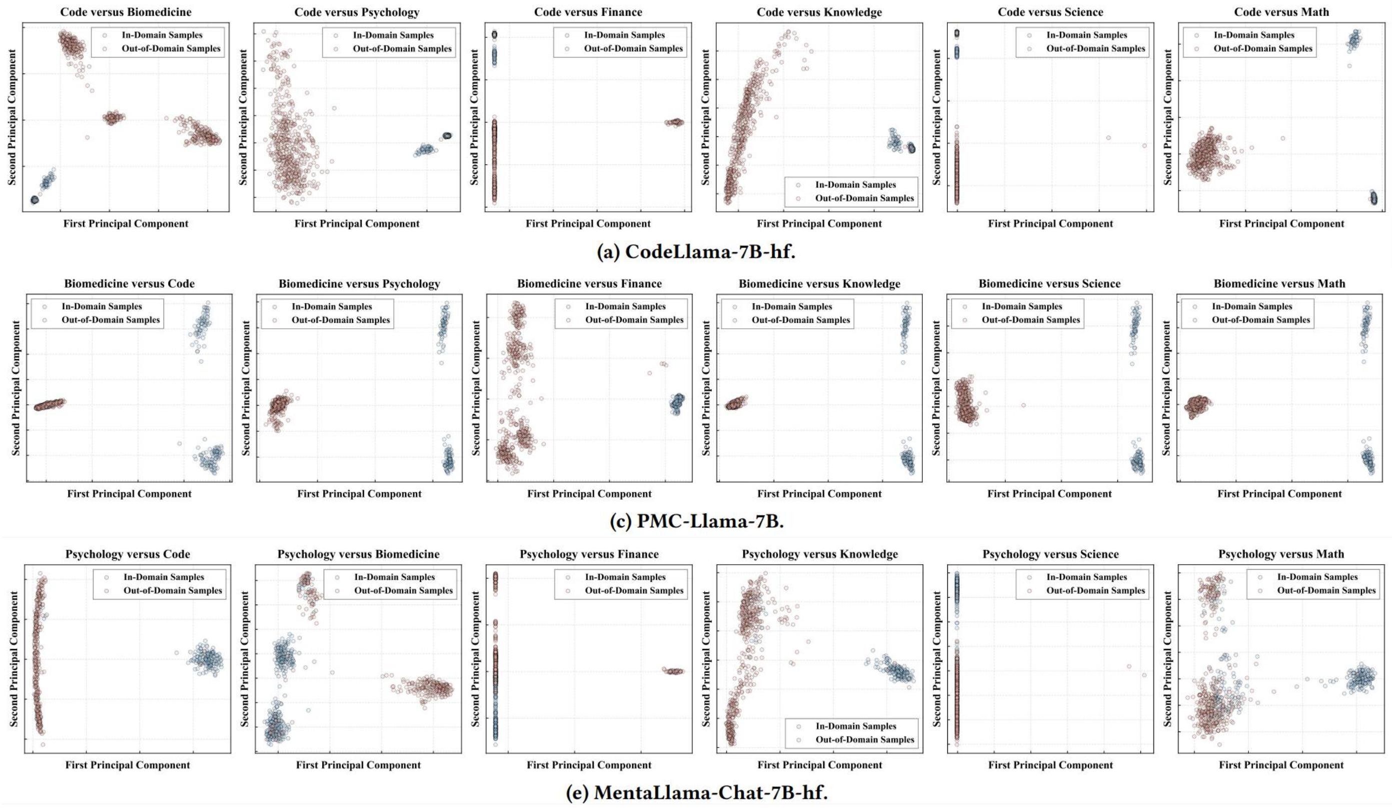
## Summary

- DomainMonitor achieves almost 100% rejection rates on out-of-domain datasets
- LLMScan presents unsatisfactory results; VALID fails to preserve effectiveness on in-domain requests

✓ DomainMonitor successfully detects out-of-domain requests; preserving in-domain performance

# Visualization

## Results



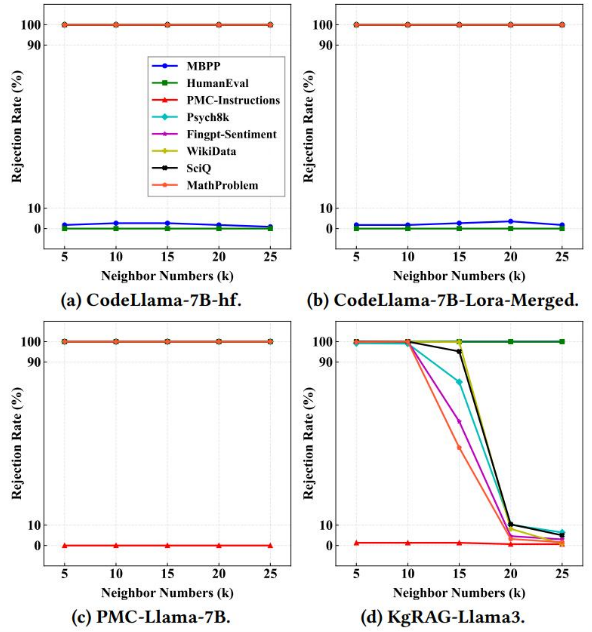

## Summary

- Utilize PCA to visualize hidden state representations on in-domain and out-of-domain requests
- ✓ DomainMonitor clearly differentiate in-domain and out-of-domain hidden states

# Ablation Analysis

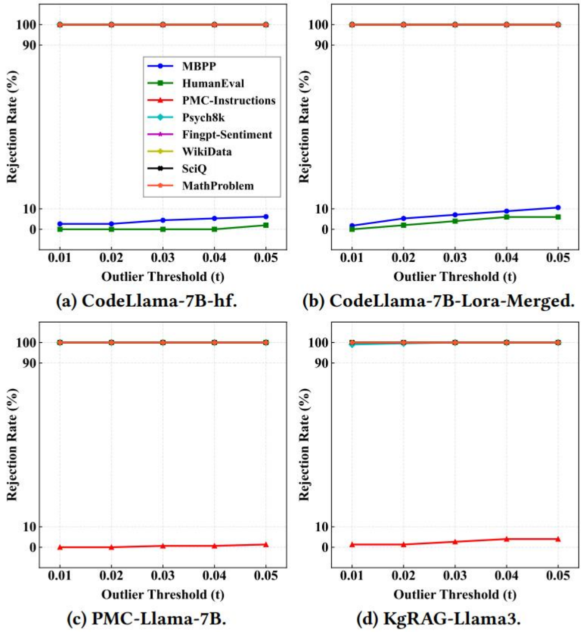
## Results

### Number of Neighbors *k* in LOF




(a) CodeLlama-7B-hf. (b) CodeLlama-7B-Lora-Merged. (c) PMC-Llama-7B. (d) KgRAG-Llama3.

### Decision Threshold *t* in LOF




(a) CodeLlama-7B-hf. (b) CodeLlama-7B-Lora-Merged. (c) PMC-Llama-7B. (d) KgRAG-Llama3.

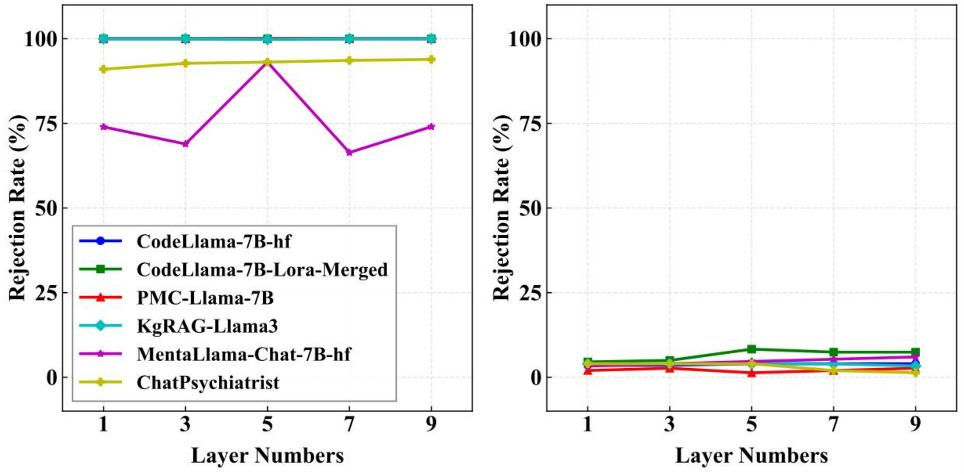
## Summary

- $k$ has almost no impact on in-domain performance; a large $k$ degrades out-of-domain performance
- $t$ has little impact on the performance of DomainMonitor
- ✓ DomainMonitor is generally stable and reliable with parameter variations

# Ablation Analysis

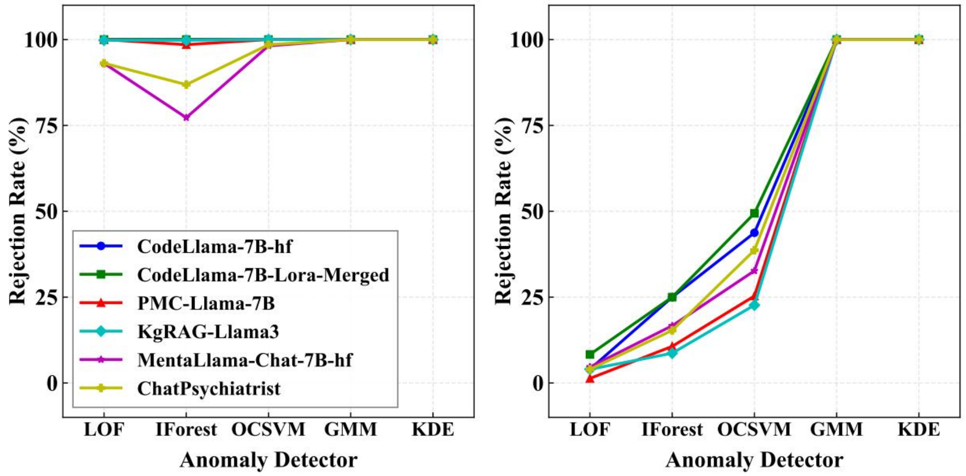
## Results

### Hidden State Layers




(a) Out-of-domain results. (b) In-domain results.

### Anomaly Detectors




(a) Out-of-domain results. (b) In-domain results.

## Summary

- DomainMonitor is stable in different layer settings, and performs best on five layers of hidden states
- LOF outperforms the other detectors, particularly in mitigating in-domain false alarms

✓ Results are consistent with our designs in DomainMonitor

# Summary

We propose DomainMonitor, the first runtime monitoring framework focusing on out-of-domain behaviors in domain-specific LLMs

DomainMonitor monitors out-of-domain requests through analyzing LLM internal hidden states regarding domain behaviors

DomainMonitor is effective in exposing anomalous hidden states, achieves promising performance on both in-domain and out-of-domain datasets

# *Thanks for Listening!*